KAIZEN MATHS

Lesson starter

Poisson Distribution

Parameters, Sums and Conditional Probability

Name _____

S1

Use the addition rule for independent Poisson variables.

Independent variables X ~ Po(3) and Y ~ Po(4). Find P(X + Y = 5).

S2

Write the conditional probability as a fraction of two probabilities.

If X ~ Po(4), find P(X = 2 | X ≤ 3).

S3

Use $P(X = 0) = e^{-\lambda}$ to find lambda.

For X ~ Po(lambda), P(X = 0) = 0.13534. Find lambda.

S4

Use the ratio of consecutive probabilities to find lambda.

For X ~ Po(lambda), $\frac{P(X=5)}{P(X=4)}$ = 0.8. Find lambda.

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KAIZEN MATHS

Exit ticket

Poisson Distribution

Parameters, Sums and Conditional Probability

Name _____

EA1

Use the addition rule for independent Poisson variables.

Independent variables $X \sim Po(3)$ and $Y \sim Po(6)$. Find $P(X + Y = 7)$.

EA2

Write the conditional probability as a fraction of two probabilities.

If $X \sim Po(2)$, find $P(X = 1 \mid X \leq 3)$.

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KAIZEN MATHS

Exit ticket

Poisson Distribution

Parameters, Sums and Conditional Probability

Name _____

EB1

Use $P(X = 0) = e^{-\lambda}$ to find lambda.

For $X \sim Po(\text{lambda})$, $P(X = 0) = 0.04979$. Find lambda.

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KAIZEN MATHS

Exit ticket

Poisson Distribution

Parameters, Sums and Conditional Probability

Name _____

EA1

Use the addition rule for independent Poisson variables.

Independent variables $X \sim Po(3)$ and $Y \sim Po(6)$. Find $P(X + Y = 7)$.

EA2

Write the conditional probability as a fraction of two probabilities.

If $X \sim Po(2)$, find $P(X = 1 \mid X \leq 3)$.

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KAIZEN MATHS

Exit ticket

Poisson Distribution

Parameters, Sums and Conditional Probability

Name _____

EB1

Use $P(X = 0) = e^{-\lambda}$ to find lambda.

For $X \sim Po(\text{lambda})$, $P(X = 0) = 0.04979$. Find lambda.

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Teacher answers & working

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S1

Use the addition rule for independent Poisson variables.

Independent variables X ~ Po(3) and Y ~ Po(4). Find P(X + Y = 5).

Answer:

$P(X + Y = 5) = 0.1277.$

Working:

Independent Poisson variables can be added by adding their parameters.

$X + Y \sim Po(3 + 4) = Po(7).$

$P(X + Y = 5) = \frac{e^{-7}7^5}{5!}.$

$P(X + Y = 5) = 0.1277.$

S2

Write the conditional probability as a fraction of two probabilities.

If X ~ Po(4), find P(X = 2 | X ≤ 3).

Answer:

$P(X = 2 \mid X \leq 3) = 0.3380.$

Working:

Use $P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$.

Here X = 2 is inside the event X ≤ 3, so P(X = 2 and X ≤ 3) = P(X = 2).

$P(X = 2) = \frac{e^{-4}4^2}{2!} = 0.1465.$

$P(X \leq 3) = 0.0183 + 0.0733 + 0.1465 + 0.1954 = 0.4335.$

$P(X = 2 \mid X \leq 3) = \frac{0.1465}{0.4335} = 0.3380.$

S3

Use $P(X = 0) = e^{-\lambda}$ to find lambda.

For X ~ Po(lambda), P(X = 0) = 0.13534. Find lambda.

Answer:

lambda = 2.000.

Working:

For a Poisson distribution, $P(X = 0) = e^{-\text{lambda}}$.

So $e^{-\text{lambda}} = 0.13534$.

Take natural logarithms: −lambda = ln(0.13534).

lambda = −ln(0.13534) = 2.000.

Teacher answers & working

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S4

Use the ratio of consecutive probabilities to find lambda.

For X ~ Po(lambda), $\frac{P(X=5)}{P(X=4)}$ = 0.8. Find lambda.

Answer:

lambda = 4.

Working:

Write the two probabilities using the Poisson formula.

$\frac{P(X=5)}{P(X=4)} = \frac{e^{-lambda}lambda^5/5!}{e^{-lambda}lambda^4/4!}$.

Cancel the common factors $e^{-lambda}$ and $lambda^4$.

$\frac{P(X=5)}{P(X=4)} = \frac{lambda}{5}$.

$\frac{lambda}{5}$ = 0.8, so lambda = 0.8 x 5 = 4.

EA1

Use the addition rule for independent Poisson variables.

Independent variables X ~ Po(3) and Y ~ Po(6). Find P(X + Y = 7).

Answer:

$\mathrm{P}(\mathrm{X}+\mathrm{Y}=7)=0.1171.$

Working:

Independent Poisson variables can be added by adding their parameters.

$\mathrm{X}+\mathrm{Y} \sim \mathrm{Po}(3+6)=\mathrm{Po}(9).$

$\mathrm{P}(\mathrm{X}+\mathrm{Y}=7)=\frac{e^{-9}9^7}{7!}.$

$\mathrm{P}(\mathrm{X}+\mathrm{Y}=7)=0.1171.$

EA2

Write the conditional probability as a fraction of two probabilities.

If X ~ Po(2), find P(X = 1 | X ≤ 3).

Answer:

$\mathrm{P}(\mathrm{X}=1 \mid \mathrm{X} \leq 3)=0.3158.$

Working:

Use $\mathrm{P}(\mathrm{A} \mid \mathrm{B})=\frac{P(AandB)}{P(B)}$.

Here X = 1 is inside the event X ≤ 3, so P(X = 1 and X ≤ 3) = P(X = 1).

$\mathrm{P}(\mathrm{X}=1)=\frac{e^{-2}2^1}{1!}=0.2707.$

$\mathrm{P}(\mathrm{X} \leq 3)=0.1353+0.2707+0.2707+0.1804=0.8571.$

$\mathrm{P}(\mathrm{X}=1 \mid \mathrm{X} \leq 3)=\frac{0.2707}{0.8571}=0.3158.$

Teacher answers & working

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EB1

Use $P(X = 0) = e^{-\lambda}$ to find lambda.

For X ~ Po(lambda), P(X = 0) = 0.04979. Find lambda.

Answer:

lambda = 3.000.

Working:

For a Poisson distribution, $P(X = 0) = e^{-lambda}$.

So $e^{-lambda} = 0.04979$.

Take natural logarithms: −lambda = ln(0.04979).

lambda = −ln(0.04979) = 3.000.